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Problems on Addition and Subtraction



Suppose a problem involving three or more numbers with a combination of addition (+) and subtraction (−) signs is given.

For solving such a problem, we proceed as per the following steps:

Step 1: Add all the numbers with + sign before them.

Step 2: Add all the numbers with − sign before them.

Step 3: Subtract the second sum from the first sum.

Note: The first number written without a + or − sign is taken up as with + sign.

Solving problems with combinations of addition and subtraction helps us solve a variety of word problems as you shall see in the examples given below:



Solved Examples

Example 1: Simplify: $3678 - 1487 + 2456 - 2798$.

Solution:

Step 1: Adding the numbers with + sign before them, we get:

Th	H	T	O
3	6	7	8
+	2	4	5
6	1	3	4

Step 2: Adding the numbers with − sign before them, we get:

Th	H	T	O
1	4	8	7
+	2	7	9
4	2	8	5



Step 3: Subtracting the second sum from the first sum, we get:

	Th	H	T	O
	6	1	3	4
-	4	2	8	5
	1	8	4	9

Thus, $3678 - 1487 + 2456 - 2798 = 1849$.



Example 2: The population of a village is 6410. If there are 1973 women and 1459 children, how many men are there in the village?

Solution: Total population of the village = 6410
 Number of women in the village = 1973
 Number of children in the village = 1459
 Number of women and children in the village
 $= 1973 + 1459$
 $= 3432$
 Number of men in the village
 $= 6410 - 3432$
 $= 2978$

Hence, the number of men in the village is 2978.

	1	9	7	3
+	1	4	5	9
	3	4	3	2
	6	4	1	0
-	3	4	3	2
	2	9	7	8

Example 3: 7016 persons visited the zoo on the first three days of a week. If 2548 and 1979 persons visited the zoo on Monday and Tuesday respectively, how many persons visited the zoo on Wednesday?

Solution: Number of persons who visited the zoo on 3 days = 7016
 Number of persons who visited the zoo on Monday = 2548
 Number of persons who visited the zoo on Tuesday = 1979
 Number of persons who visited the zoo on Monday and Tuesday
 $= 2548 + 1979$
 $= 4527$
 Number of persons who visited the zoo on Wednesday
 $= 7016 - 4527$
 $= 2489$

Hence, 2489 persons visited the zoo on Wednesday.

	2	5	4	8
+	1	9	7	9
	4	5	2	7
	7	0	1	6
-	4	5	2	7
	2	4	8	9



Exercise 13

Simplify:

1. $3716 - 1829 + 697$
2. $2368 - 3459 + 2987$
3. $2513 - 589 + 732 - 1067$
4. $6823 - 1946 + 2761 - 1250$
5. $5701 + 642 - 1396 - 2381$
6. $8300 + 1032 - 5431 - 2357$
7. Subtract the sum of 2999 and 3888 from 9111.
8. Subtract 4235 from the sum of 3672 and 1893.
9. Subtract the difference of 3672 and 1893 from 4235.
10. Subtract the difference of 4587 and 6375 from the difference of 7354 and 3689.
11. The sum of three numbers is 9010. If two of them are 3768 and 1697, find the third number.
12. Gopal had ₹ 7520. He purchased a chair for ₹ 1765 and a table for ₹ 2978. How much money was left with him?
13. From a rope 3000 m long, two pieces measuring 876 m and 1057 m were cut off. Find the length of the remaining rope.
14. Mrs Janki needed ₹ 9000 to buy a necklace. She had ₹ 5436 with her and took a loan of ₹ 3349 from her friend Geeta. How much is she still short of?
15. An orchard has 5304 trees. There are 1647 coconut trees, 2798 mango trees and rest orange trees. How many orange trees are there in the orchard?
16. Amit had to travel 4325 km. He travelled 2678 km by train, 1354 km by bus and the rest by car. What distance did he travel by car?
17. The population of a town is 8215. If there are 3438 men and 2859 women, how many children are there in the town?

